

End-to-End Data Management Pipeline for a Recommendation System

RecoMart Recommendation System

Team Member Details: Student Group / RecoMart Data Platform Team

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1. Problem Statement

RecoMart wants to create a scalable recommendation engine that uses clickstream, purchase, and product metadata to provide personalized product suggestions and improve engagement and conversion rates.

2. Objectives

- Ingest interactions and product data from multiple sources
- Validate data quality and monitor issues
- Prepare and transform data for downstream modeling
- Build a feature store and train a recommendation model
- Orchestrate the pipeline with Prefect and track experiments with MLflow

3. Methodology / Pipeline

The implemented workflow follows ingestion -> validation -> preparation -> feature engineering -> feature store -> model training.

4. Implementation Details

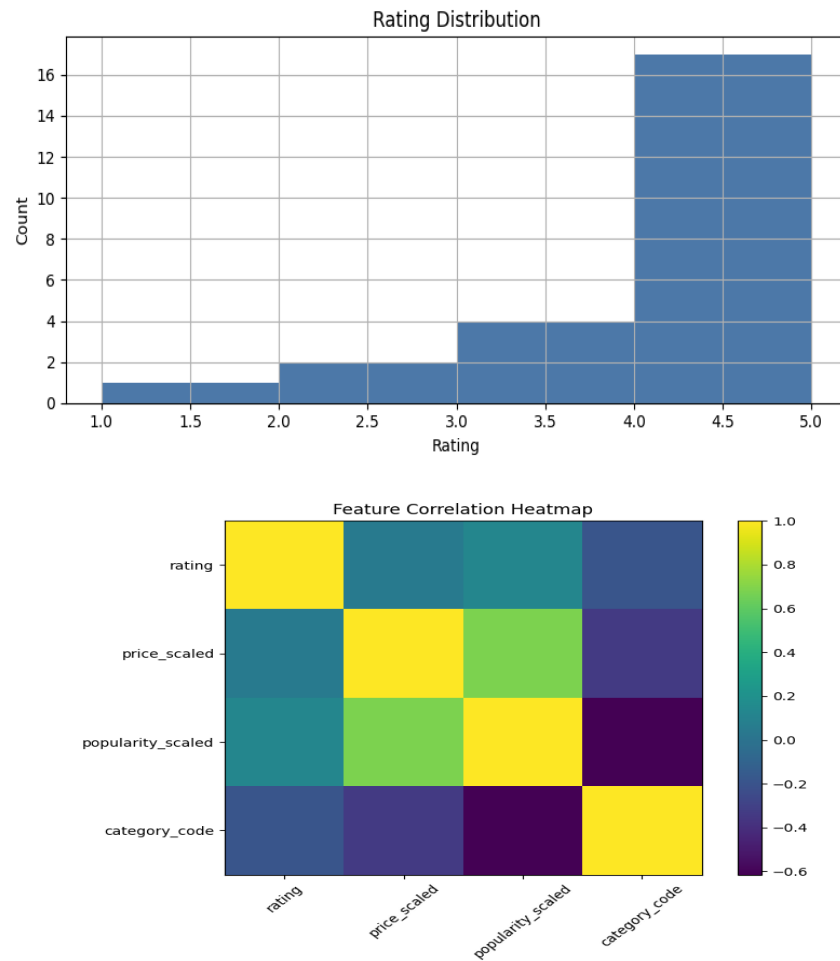
- Ingestion scripts read local CSV and JSON data into a structured raw-data lake layout.
- Validation checks verify missing values, duplicate rows, and rating ranges.
- Preparation joins product metadata with interaction data, encodes categories, and normalizes attributes.
- Feature engineering creates user activity frequency, average ratings, and popularity-based features.
- Prefect orchestrates the workflow while MLflow records parameters, metrics, and artifacts.

5. Results and Outputs

Metric	Value
Model	NMF
NMF Components	3
RMSE	1.9582
Precision@5	0.0

Validation summary: 2 datasets checked, passed = True

Sample plots generated:



6. Conclusion and Future Scope

The pipeline provides a modular and reproducible foundation for recommendation-system data management. Future work can extend it to near-real-time ingestion, richer feature engineering, and deployment via an inference API.

7. Submission Links

Google Drive Link to Video Walkthrough: [Add link here]

Google Drive Link to Deliverables ZIP: [Add link here]

Repository: <https://github.com/sharnika55/DMML-Assignment>